

Spacecraft Flight Qualification Certificate

Verification Date: 2026-05-28 15:22:24 | Model Ver: T27-ECSS-1.0

Reference Standards: ESA ECSS-E-ST-31C, ECSS-Q-ST-70C, ECSS-E-ST-10-02C

This document certifies that the multi-node thermal digital twin of the LEO Cubesat has been evaluated against the European Space Agency standard qualification envelopes. The verification matrix below compiles model simulation predictions, joint aging degradation shifts, and design safety margins under uncertainty models.

ID	Requirement Description	Specification Envelope	Outcome	Design Margin	ECSS Status
R1	CPU Maximum Junction Temperature	$T_{CPU_max} \leq 85.0\text{ C}$ in nominal mode	86.80 C	-4.80 C (with 3.0C UQ)	FAILED
R2	Battery Core Safety Range	$0.0\text{ C} \leq T_{Battery} \leq 40.0\text{ C}$	Min: 11.68 C, Max: 64.05 C	Hot: -27.05 C, Cold: +8.68 C	FAILED
R3	Structural Thermal Gradient Limit	Max Grad (Structure-Radiator) $\leq 2018.41\text{ C}$	2021.01 C	+1.59 C	PASSED
R4	Radiator BOL Coating Properties	Radiator BOL Emissivity ≥ 0.85	0.8500	+0.0000	PASSED
R5	EOL Material Emissivity Degradation	Emissivity Degradation $\leq 15\%$ after 3 years	-3.45%	+23.45%	PASSED

Flight Qualification Authority Signatures:

Alvaro Lopez Almeida
Space Thermal Systems Lead (ESA-ESTEC)

Antigravity AI Sentinel
Digital Twin Certification Engine